VISWAJIT R

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PROFESSIONAL SUMMARY

AI & Data Science undergraduate with a strong foundation in machine learning, reinforcement learning, and full-stack development. Passionate about building AI-driven applications, conducting applied research, and developing scalable software solutions. Experienced in leading technical teams, delivering production-ready web applications, and transforming research ideas into practical systems.

EDUCATION

Nitte Meenakshi Institute of Technology (NMIT), Bengaluru – 2023 - 2027
4th Year, B.E. in Artificial Intelligence & Data Science
CGPA: 9.22 / 10

Sri Chaitanya Techno School (SCTS), Bengaluru – 2021 - 2023
Class XII (CBSE)
Percentage: 86.4%

New Baldwin International School (NBIS), Bengaluru – 2018 - 2021
Class X (ICSE)
Percentage: 97.6%

EXPERIENCE

Technical Head — AUGMENTIX (AI&DS Student Club) Nov 2024 – Present
NMIT, Bengaluru

- Authored and presented a research paper on AI-driven stock market price prediction to an audience of 50+ students and faculty
- Led 12 junior developers across 3 incubation projects, maintaining 100% milestone completion
- Co-developed an AR/VR curriculum with Madras Mindworks used by over 80 engineering students
- Developed the official HackSurgeX website supporting registrations for a 36-hour hackathon with 500+ visitors and supported logistics for events like NirvAI 2024 and Hallomania 2025 attracting 250+ participants

CERTIFICATIONS

- NPTEL – Fundamentals of Object-Oriented Programming |*Percentage - 94%*
- Microsoft – Java Full Stack Developer
- IITM BS Degree Data Science Foundational Certificate
- Simplilearn – Introduction to MS Excel
- Simplilearn – Data Science Foundation Certificate

PROJECTS

Causal Reinforcement Learning Framework for Algorithmic Trading 2025 – Present

Final Year Project | Python, Causal Inference, Reinforcement Learning

- Developed a causal reinforcement learning framework combining NOTEARS and Structural Causal Models for algorithmic trading across 30 financial assets.
- Improved Sharpe Ratio over non-causal baselines using Dueling Double DQN and Discrete SAC.
- Validated results with statistical significance testing and 7-model ablation study.

Document Forgery Detection — Deep Learning Model Comparison | Github May 2026

Academic Project | Python, TensorFlow/PyTorch, CNNs

- Implemented and evaluated five deep learning architectures (Custom CNN, EfficientNetB0, ResNet50, MobileNetV2, and a hybrid model) to classify forged/manipulated documents
- Benchmarked five CNN architectures using accuracy, precision, recall, and F1, identifying the best-performing architecture for the task

RhythmX — Algorithmic Music Discovery Platform | Github June 2024

Academic Project | React, FastAPI, Python, MongoDB, scikit-learn

- Developed a full-stack music recommendation system clustering songs by audio-feature "mood" (energy, valence, tempo) using K-Means and Librosa-based feature extraction
- Built a FastAPI backend applying association-rule mining (Apriori) on community listening patterns to recommend niche tracks outside mainstream popularity signals, with a React frontend

HackSurge X — Hackathon Event Landing Page | Github | Live Demo Apr 2025

Personal Project | HTML, CSS, JavaScript

- Designed and built an animated, framework-free event landing page featuring a hero section, live countdown timer, and event timeline, using Canvas-based particle effects and scroll-triggered animations
- Delivered a fully responsive, cinematic UI (animations, custom cursor interactions) optimized for performance without relying on external frameworks

Bank ATM Simulation System — C++ | Github Jan 2025

Academic Project | C++, Data Structures & Algorithms

- Designed a console-based banking simulation applying object-oriented principles and core data structures (stacks, queues, vectors) to model account registration, authentication, and fund transfers
- Implemented business-rule logic for minimum-balance enforcement and service charges across dual account types (savings/current)

AI Mentor — Gemini API-Powered Student Guidance System | Github May 2024

NMIT Hacks | HTML, CSS, JavaScript, Python

- Designed and built the complete frontend for a student mentoring platform; integrated with Gemini API to deliver 24/7 personalized academic guidance
- Prototyped an AI mentoring system using the Gemini API in a 48-hour hackathon, providing adaptive, context-aware student support at scale

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Go, MATLAB

AI/ML: Machine Learning, Deep Learning, Reinforcement Learning (DQN, SAC), Causal Inference (PC Algorithm, NOTEARS, Structural Causal Models), Computer Vision & Digital Image Processing (CNN, ResNet50, EfficientNetB0, MobileNetV2), Clustering (K-Means), Association Rule Mining, LLM API Integration (Gemini API)

Frameworks & Libraries: PyTorch, TensorFlow, FastAPI, React, scikit-learn, Librosa, OpenCV

Web & Databases: HTML, CSS, JavaScript, Web Technologies, MongoDB, SQL, Excel, PowerBI, Power Automate, Git

Practices: Object-Oriented Programming, Data Structures & Algorithms, Agile/Scrum, Technical Team Leadership, Event Management, Curriculum Development